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APPLICATION - SOFTWARE DEVELOPMENT ENGINEER

Dear Esri Team,

What particularly interests me about this position is the opportunity to work on indoor positioning at the intersection of C++ software development, localization, sensor fusion, and spatial data. In autonomous robotics, I have worked extensively on estimating position and motion from sensor data and using this information for spatial mapping and navigation. Applying this experience to ArcGIS IPS particularly interests me.

At Baubot, I took technical responsibility for building a ROS 2 software stack from scratch in C++, including autonomous localization and navigation. On the physical robot, I worked on correcting odometry drift using external total station measurements. I also developed multithreaded C++ interfaces with Python APIs for easier use by the team and tested the complete software stack end-to-end in headless Gazebo simulations as part of the CI/CD pipeline.

My academic work further focused on localization, state estimation, and spatial representations. I worked with methods such as AMCL and Kalman filtering, including Kalman-based motion estimation in my Bachelor's thesis. In my Master's thesis, I developed an autonomous exploration system combining RGB-D perception with persistent 3D semantic mapping of unknown environments for semantic object search through autonomous navigation.

I am particularly interested in transferring this background to indoor positioning on mobile devices, where combining motion information with different sensor sources presents similar challenges in localization and state estimation. At Esri, I would like to contribute this experience while further developing my knowledge of indoor positioning technologies.

I look forward to discussing how my background in C++, localization, state estimation, and spatial mapping can contribute to ArcGIS IPS. Further details on my work and projects can be found in my [portfolio](#).

Kind regards,



Kevin Eppacher